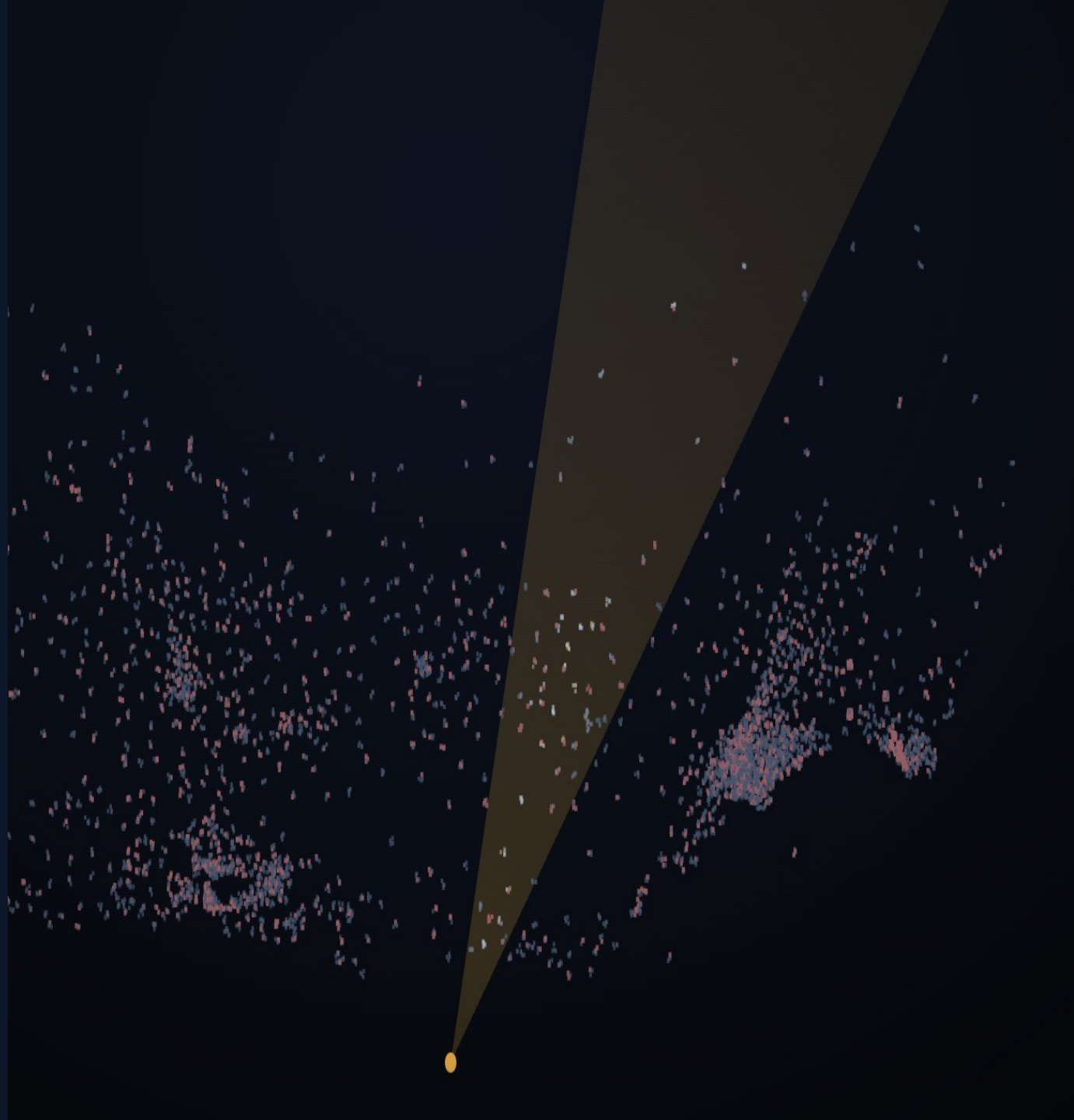
DataNova 2026 · Problem statement DN-BU-01

Lighthouse

Churn prediction and causal retention planning

See who is about to leave, what each customer is worth, and the offer that will actually keep them.



Team Tiramisu Tensor

Live app: lighthouse-retention.vercel.app
Videos: lighthouse-retention.vercel.app/watch
Code: github.com/ac265640/LightHouse

The problem: customers leave, and teams can't tell who to save

Our problem statement

DN-BU-01: Customer Churn Prediction and Retention Strategy

What the brief asks for:

- A churn prediction model
- Insights about customer segments
- A retention playbook the business can act on

Our data: IBM Telco Customer Churn 11.1.3, 7,043 customers of a California telecom, 52 columns covering services, billing, offers and exit reasons.

What churn cost this company in one quarter

26.5%
of all customers left in a single quarter

1,869
customers lost, out of 7,043

$139k
of monthly revenue walked out the door

$389k
of lifetime value still at risk among those who stayed

Why most churn projects never pay off

A high accuracy score is where the hard questions start. Three traps sit between a good model and a retention program that makes money.

Models that cheat

Some columns are only recorded after a customer leaves. Keep them and the model looks perfect in testing, then fails in real use.

0.998

accuracy (AUC) of a model that cheats

Risky is not the same as persuadable

The riskiest customers often leave whatever you offer, and many others would have stayed anyway. Ranking by risk wastes the budget.

−$11.5k

result of the usual approach on a $15k budget

No sense of time or value

A yes-or-no score can't say how many months a customer has left, so it can't price a save or say who to call first.

1 quarter

of data: time has to be modelled, not assumed

Our solution: Lighthouse answers three questions



Who is leaving, and why?

Five model families compete, then a calibrated ensemble scores every customer. SHAP shows the top five reasons for each one.



When, and what are they worth?

A survival model estimates the months each customer has left, and turns that into lifetime value and value at risk.



What will actually keep them?

Causal inference measures how much each offer really changes churn, for each customer. A budget optimiser then picks who to contact.



The output: a ranked call list

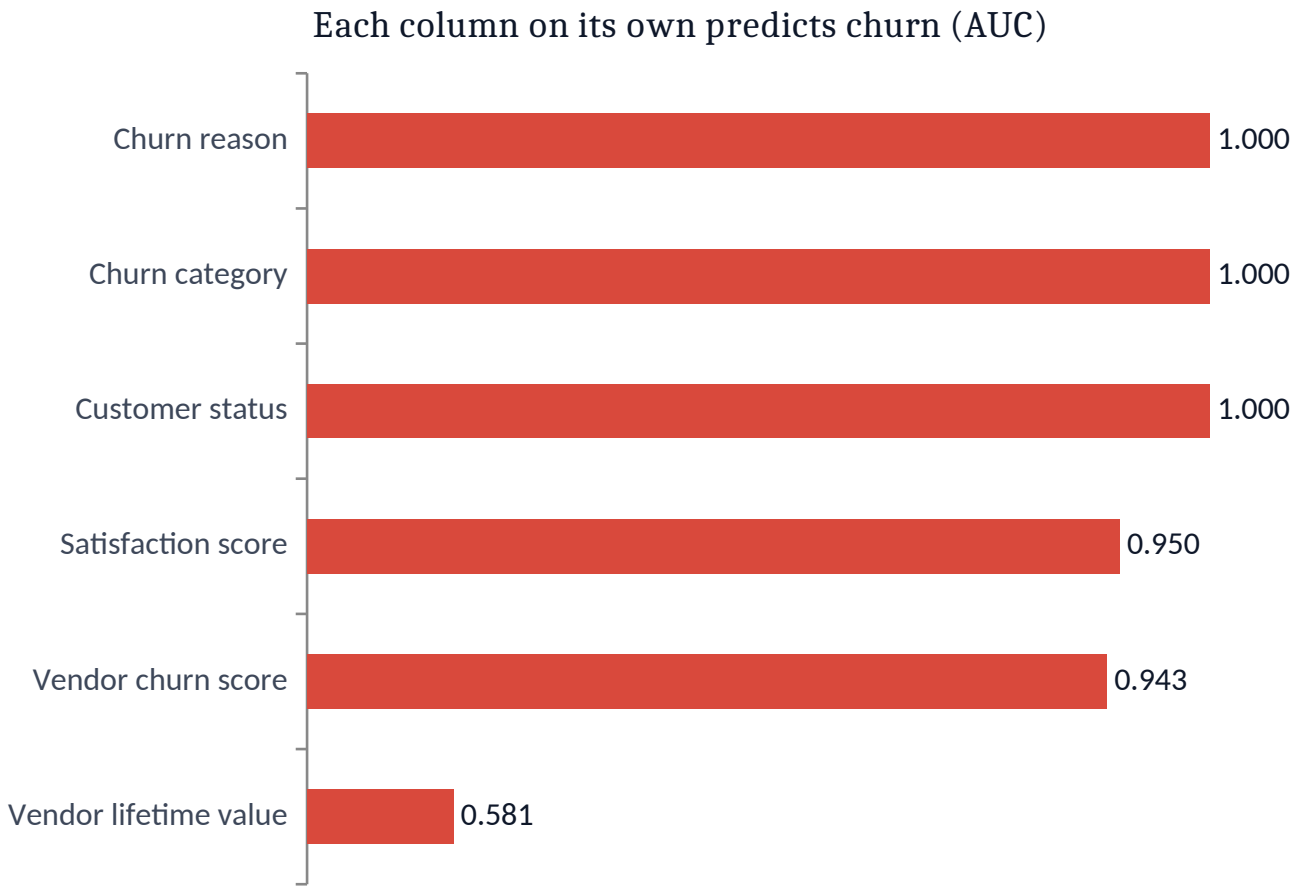
Who to contact, with which offer, and the expected return for each.



Delivered as a live product, not a notebook

A web workspace for retention analysts, an API that scores new customers, and a Gemini AI copilot that answers questions from the real numbers.

Step 1: we refused to let the model cheat



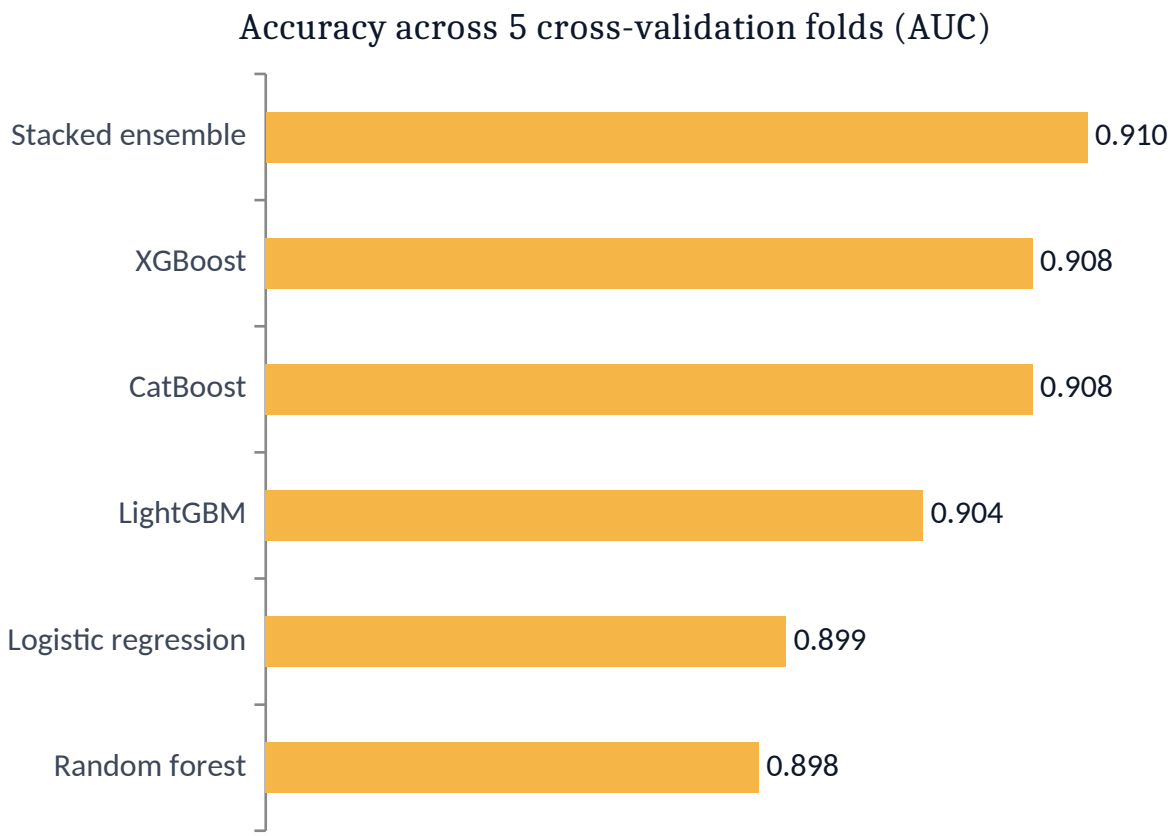
We tested every column alone. These six are only written after a customer leaves, so we removed all of them.

0.998
with the leaks: looks perfect, fails in use

0.923
honest score on unseen customers

- Gender is kept out of the model and used only to audit fairness. It added just 0.001 AUC.
- 10 new signals were engineered, such as bill drift, refunds and services held.

Step 2: five models compete, the best is calibrated



The stacked ensemble combines all five, then we calibrate it so its probabilities are honest.

0.923

accuracy (AUC) on 1,409 unseen customers

1.4%

calibration error: a 30% score means 30 in 100 leave

3.4×

more churners in the top 10% than average

93%

of customers get a reliable confidence label

Step 3: how long each customer has, and what they're worth

Our data is a one-quarter snapshot, so everyone in it has already survived to today. We corrected for that with a survival model (Cox, left-truncated), so every customer gets a timeline and a dollar value.

Before the fix

87% risk this quarter, yet 52 months left. A contradiction.

After the fix

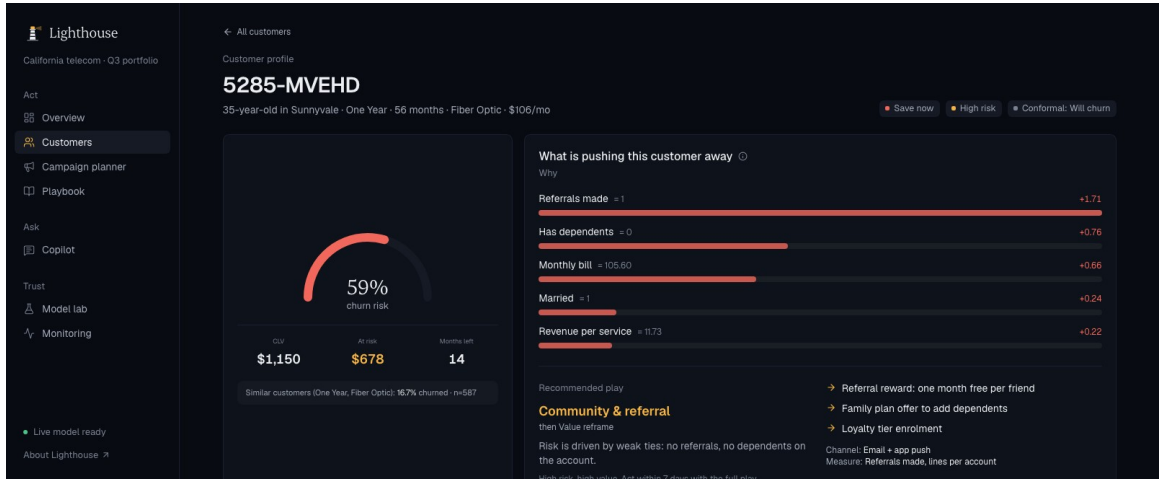
Two separate models now agree on who is at risk (correlation 0.95).

0.83

survival model accuracy (c-index)

0.90

accuracy predicting this quarter's churn



Example: customer 5285-MVEHD

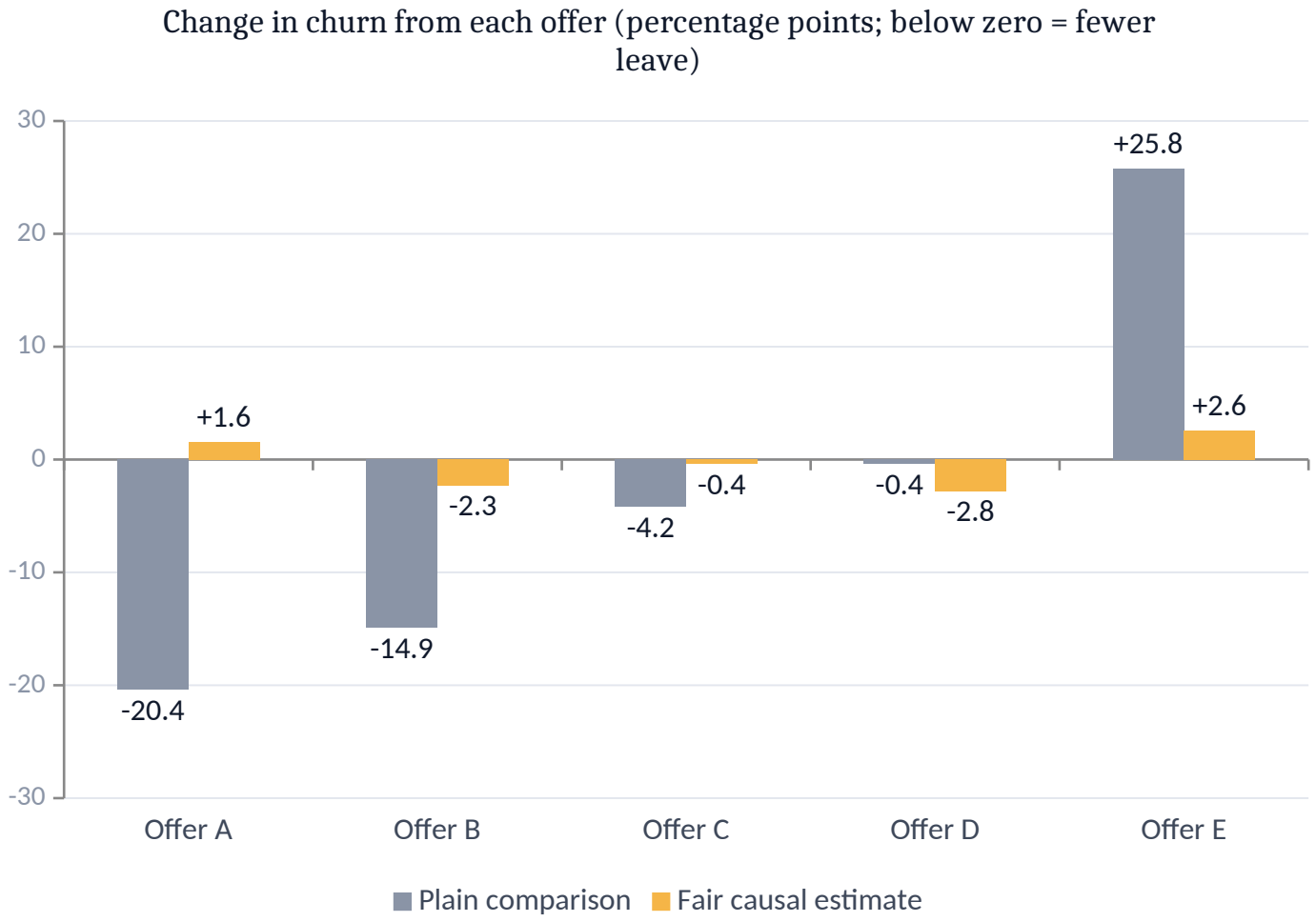
59% churn risk

14 months left

$1,150 worth if kept

$678 value at risk

Step 4: the offer that looked best did nothing



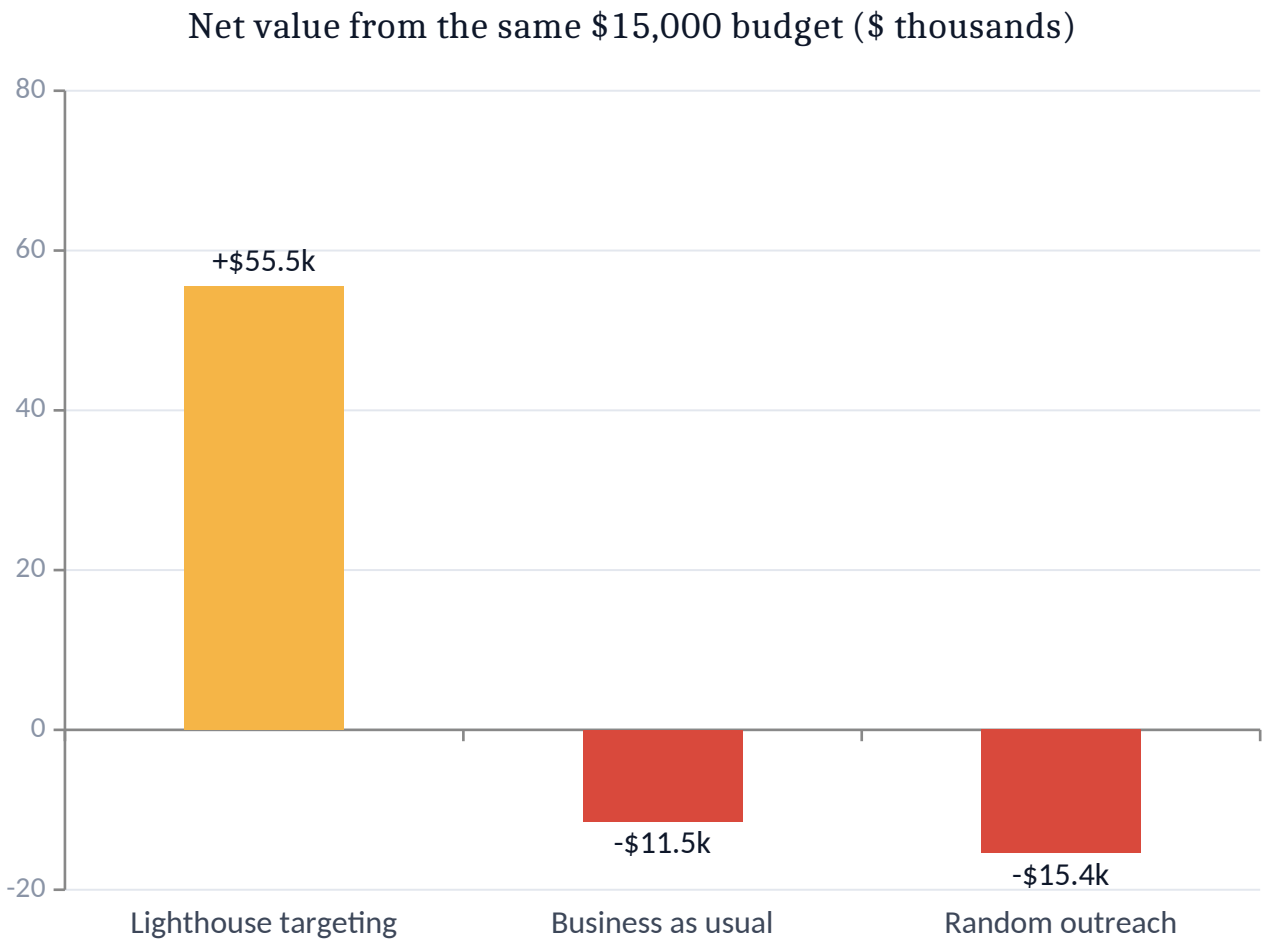
The twist

A plain comparison says Offer A cuts churn by 20 points. But Offer A only goes to customers who have stayed 66+ months, and they rarely leave anyway.

What we found

- Compared fairly (doubly-robust, inside each tenure tier), Offer A's effect is +1.6 points.
- No offer works on average. The value is in choosing who receives it.
- Every campaign comes with an A/B test size to confirm it.

Step 5: spend the budget where it changes the outcome



3.7×

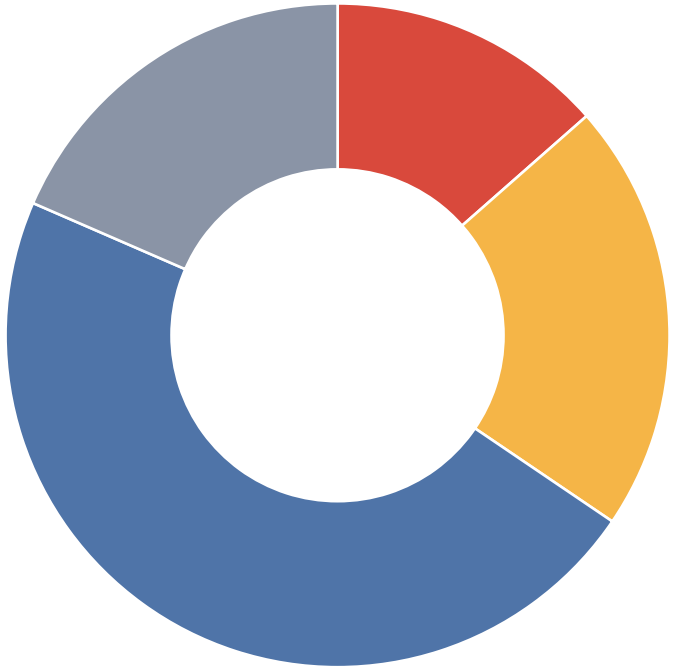
return on the campaign budget, where the usual approach loses money

How the optimiser decides

- For each customer and offer: churn prevented × lifetime value − offer cost
- Rank everyone, then spend the budget from the top
- Result: 400 customers contacted, about 65 saved

Segments and the retention playbook

5,174 active customers by priority



- **Save now** 699
- **Low-cost save** 1,085
- **Grow** 2,434
- **Maintain** 956

Every at-risk customer gets a play, chosen from their own churn drivers

Contract lock-in

Discounted 12 or 24-month contract, priced from their value

1,227 customers

Community & referral

One month free per friend referred, family plan to add lines

444 customers

Early-life onboarding

Welcome call and setup help in the first 90 days

55 customers

Value reframe

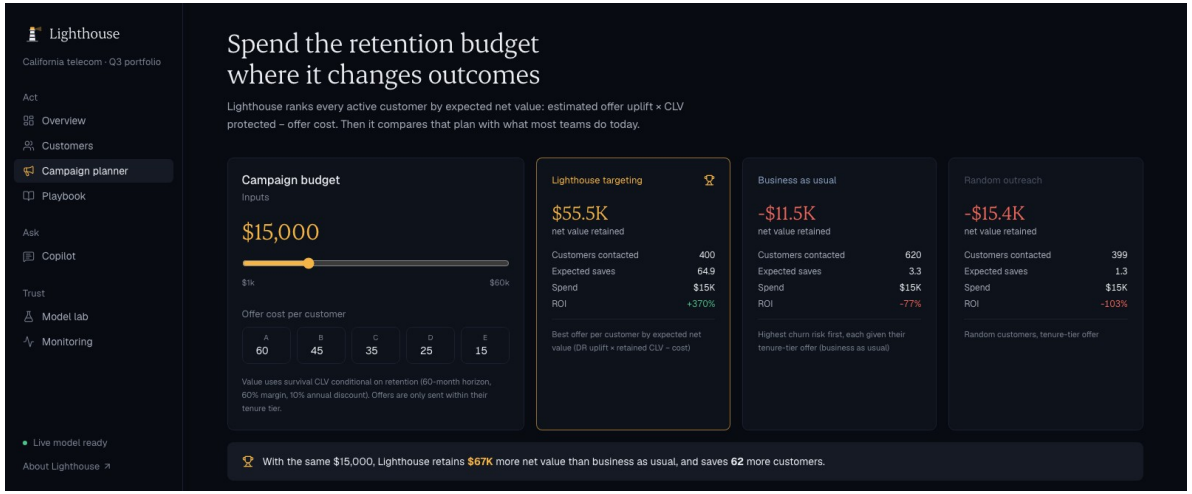
Plan right-sizing and bundles when the bill is the problem

38 customers

Seven plays in total, each with actions, a channel and a KPI to track.

The product: a live workspace for retention teams





Live customer map

Every customer at their real location, lit by risk

Customer profile + what-if

Change a contract and watch the risk move: 59% → 29%

AI retention email

Gemini writes a personal email from each customer's drivers

Campaign planner

Budget split, three strategies, A/B test size, CRM export

AI copilot

Answers from Lighthouse's own data and shows its sources

Model lab + monitoring

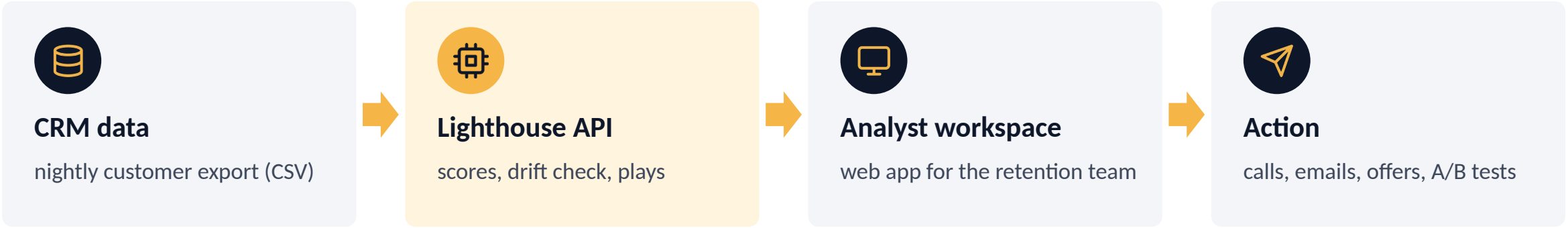
Evidence, fairness audit and drift checks on new data

Try it: lighthouse-retention.vercel.app

How Lighthouse is different

	A typical churn project	Lighthouse
Testing	Random split, leaky columns left in	Leakage audit and a locked holdout of unseen customers
Probabilities	Scores that only rank customers	Calibrated: 30% means 30 in 100, plus confidence labels
Time and value	Will they leave: yes or no	Months left and lifetime value for every customer
Offers	Assume the offer works	Causal estimate of what each offer really changes
Decision	A list of the riskiest customers	A budget plan with expected return and an A/B test size
AI	A chatbot that can guess numbers	A copilot that only uses numbers from the model's tools
Delivery	A notebook	A live web app and API, 30 automated tests, $0 hosting

Built to run in a real company



Rollout plan

Weeks 1–2

Score the nightly export

Connect the CRM file, check drift first, send call lists to the team

Months 1–2

Prove the offers

Run the A/B tests the planner sizes, retrain the offer model on the results

Quarter 2

Close the loop

Trigger plays automatically when a customer's risk crosses the threshold

Works for any subscription business telecom, streaming, SaaS, insurance, banking: anything with tenure, billing and offers. Runs today on free hosting.

Built to be trusted, honest about its limits

Fairness audit

Recall is checked by gender, age and seniority. Gaps are shown in the app, not hidden, and gender is never a model input.

Drift monitoring

Every feature is compared with training data. New customer files get a stable, watch or drift verdict before anyone acts.

Tested and reproducible

30 automated tests, one command to retrain every model, and an AI copilot with read-only access and usage limits.

What we are honest about

- One quarter from one company: retrain on a client's own history before use.
- Offer effects are observational and none is significant alone, so every campaign is A/B tested.
- Exit reasons can't be predicted from the profile (44% vs a 45% baseline), so plays use churn drivers instead.
- Recall differs by group; the fairness audit makes it visible.

Lighthouse turns a churn score into a plan

Who to save, what they're worth, and the offer that will actually keep them.

0.923	1.4%	0.83	3.7×
honest accuracy (AUC)	calibration error	survival accuracy	campaign return

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Code github.com/ac265640/LightHouse

Thank you

Team Tiramisu Tensor

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